

Site: Americas	Procedure Type: AP	System: Safety	Version/Issue Date: V7, 2/15/2026
Review By Date: February 15, 2027			
AP Overview: Documented safety guidelines and expectations for General Contractors building or retrofitting STACK Infrastructure data centers.			

Table of Contents

1.	Purpose.....	2
2.	Scope.....	2
3.	Applicable Contract-Required Plans/Documents	2
4.	Detailed Processes and Policy	2
5.	General Standards	3
6.	Contractor's Environment, Health, and Safety Management Plan.....	4
7.	Site Management and Rules	5
8.	High Risk Activity (HRA) Planning.....	8
9.	Safe Systems of Work.....	10
10.	Standard Work Procedures	20
11.	Safety Implementation and Administration.....	23
12.	Exemptions.....	26
13.	References	26
14.	Definitions.....	27

1. **Purpose**

This document establishes minimum environment, health, and safety (EHS) expectations for STACK Infrastructure's (STACK or Owner) general contractors. This document defines the guidelines and procedures for implementation of general contractor activities at STACK Infrastructure sites in a manner that protects people and assets from hazards.

This document may be updated periodically to reflect changes in regulations, laws or in industry best practices or business needs. Updates will be communicated in writing and by replacing this Guideline with the updated version. The latest published version of this document supersedes all prior versions and is binding on all general contractors.

2. **Scope**

This applies to general contractors or prime contractors, hereby known as the 'Contractor,' working for STACK Infrastructure.

This document sets out STACK's minimum EHS expectations for work performed by the Contractor at STACK sites. Where the Contractor's or local legislative standards are more stringent, the most stringent standard applies.

These guidelines also apply to any trade partner and their tiered trade partners working for STACK or the Contractor. It is the responsibility of the Contractor to include this document in any contracts with trade partners and their tiered trade partners, to ensure the expectations set forth in this document are implemented and enforced. The guidelines apply to all personnel under the Contractor's responsibility such as trade partners, suppliers, vendors, and visitors.

3. **Applicable Contract-Required Plans/Documents**

The following site-specific plans should be developed by the Contractor:

- Site-Specific Safety Plan
- High Risk Activity Management Plan
- Site Mobilization Plan
- Emergency Response Plan
- Traffic Management Plan
- Electrical Safety Plan
- Control of Hazardous Energy Plan
- Blasting and Explosives Site Plan
- Respiratory Protection Plan

Additional documents, when applicable to scope may be required. Some examples are Commissioning Plan, Ground Disturbance Plans, Mold Remediation, Quality Control and Protection from Falling Objects Plans.

These plans will state how workers will be protected, instruction, training and supervision and will cover all aspects of compliance. Periodic inspections and training verification of all trade partner activities is the responsibility of the General Contractor and findings of those audits will be made available to the Owner.

4. **Detailed Processes and Policy**

A formal trade partner selection process is required using key criteria to evaluate safety performance and programs, and which includes a thorough review of written safety programs and site-specific safety plans prior to contract award and mobilization.

The Contractor is to perform work in the safest manner possible, consistent with good construction practices, and is to ensure that accidents are prevented through planning, training, follow-up, and a cooperative effort in all areas of construction. Effective preventive measures should be identified and applied prior to the execution of work.

The Contractor must have a structured, systematic approach for continual improvement to eliminate the root cause of deficiencies. The Contractor and their trade partners are responsible for executing the project work with the highest level of quality and to complete the scope of work on schedule and within budget. At no time will EHS performance and considerations be compromised for quality, cost, or schedule.

5. **General Standards**

a) Overview

The Contractor acknowledges the Owner's values and strong commitment to health and safety and affirms that it has a written EHS management policy that has been signed and is actively supported and endorsed by the Contractor's management. The Contractor confirms that its written policy is widely disseminated

and understood by its employees and that it includes a description of the Contractor's organization, its safety procedures, and how it communicates with its personnel and trade partners.

b) Legal Requirements

The Contractor is to understand and comply with all applicable local, state, provincial, and federal regulations applicable to the services to be provided. The Contractor assures that their personnel and their trade partners are knowledgeable of all such laws, regulations, guidelines, and standards, and are trained in accordance with those requirements.

The Contractor further confirms that it has read the EHS guidelines outlined in this document in full and has included adequate budget and schedule for implementation and management of all EHS requirements for the duration of the contract.

The Contractor is solely responsible for carrying out the work under the Contract having the highest regard for the health and safety of its employees, trade partner's employees, and persons at or in the vicinity of the site, temporary work, materials, the property of third parties, and any work relating to the Contractor carrying out its obligations under this Contract.

c) EHS Alignment Meetings

After the award of the Contract and prior to commencing work (including mobilization and site set-up activities), as owner, STACK will hold alignment meetings by phase of construction, to ensure readiness of Contractor and trade partners associated with that phase of construction. The senior project and site management personnel, including the EHS manager, are expected to attend. These phases are:

- i) Civil / Mass Grading/Site Prep
- ii) Vertical work
- iii) Mechanical/Interior finishes
- iv) Cx/Energization
- v) Post RFS/DHFO

The Contractor will hold a kick-off EHS meeting with the Owner to:

- i) Align the contents of the Contractor's site-specific safety plan and this document and ensure

- that all applicable legal requirements have been included
- ii) Facilitate a consistent approach to EHS issues
 - iii) Ensure specific EHS risks are addressed prior to the work commencing

The Contractor's Project Manager, Project Sponsor or equivalent, and the Construction Manager should attend the kick-off meeting. Trade partner key personnel may be included in the Contractor's kick-off EHS meeting, or scheduled as a separate meeting, with the same intent and purpose.

6. **Contractor's Environment, Health, and Safety Management Plan**

a) Site-Specific Safety Plan

The Contractor will prepare, implement, and administer a site-specific safety plan within 30 days of contract award.

The plan should be in writing and submitted to the Owner for review and acceptance, prior to mobilization to the construction site and before any work commences. The site-specific safety plan will, at a minimum, meet the expectations in this document and meet applicable legal standards. Any proposed amendments or revisions to the Contractor's site-specific safety plan should be submitted to the Owner for review.

The site-specific safety plan is to provide a systematic method of managing hazards according to the risk priority, including all mobilization and site set-up activities. Owner will have the opportunity to review the plan periodically throughout the project.

The Contractor's site-specific safety plan is to be a living document, updated whenever necessary to adapt to the change of the organization, site conditions, requirements, etc. The Contractor's site-specific safety plan will address any current or pending pandemic occurrences.

b) Trade Partners

The Contractor is to ensure that all their trade partners comply with the Contractor's site-specific safety plan.

c) Off-site Pre-Fabrication

The Contractor's site-specific safety plan is to consider all the construction activities, whether performed at the site or off-site, including the Contractor's trade partner where pre-fabrication, skidding, and/or commissioning services are performed in connection with the Contractor's contracted services with the Owner.

The Contractor is responsible for initiating, maintaining, and supervising all safety precautions and programs in connection with the performance of the work. While this specification, and/or the relevant MSA, does not dictate construction means and methods or sequences of work, the Contractor should evaluate off-site safety and will be responsible for the safety while work is being performed for the project.

d) Owner Furnished, Contractor Installed (OFCI) Requirements

The Contractor's site-specific safety plan is to account for all the construction work, including oversight of the Owner's contracted vendors, commonly known as an OFCI. While on-site or at an off-site facility the Contractor is to coordinate all work activities. The Contractor is to ensure that all OFCI vendors conform to the Contractor's site-specific safety plan and all its requirements.

7. Site Management and Rules

a) Security

Subject to the Owner's standard security protocols and procedures with respect to operating facilities, the Contractor is to control access to and from the construction site and, wherever possible, fence the site. Only people and vehicles with authorization should be allowed to enter the construction site. The Contractor is to establish demarcation boundaries with appropriate signage for PPE requirements to enter the construction zones.

The Contractor will erect and maintain safeguards for the protection of workers and the public, as well as operational employees of the Owner, where applicable.

The Contractor will manage all foreseeable hazards created by performance of the work by:

- i) Providing and taking all measures necessary, in the circumstances, to maintain personal health and hygiene, ensuring safety of people and property, and protecting the environment at or near the site.
- ii) Avoiding unnecessary interference with the passage of people and property at or near the site, with specific reference to operational areas of the Owner, where applicable.

When construction projects are carried out within areas managed by STACK Security and/or Operations, the Contractor will, at a minimum, include the following in its site-specific safety plan.

- i) A deliveries process, which will include all Vendor/OFCI Equipment suppliers. The process will include means for managing 'off' hour deliveries. A schedule of deliveries will be provided to security as part of the process.
- ii) A description of site hours, to include communications of unplanned closures i.e., due to inclement weather.
- iii) A requirement that all delivery companies be made aware of the site PPE rules.
- iv) Where applicable, require delivery companies to provide equipment (forklift, lulls etc.) training records to the Contractor prior to accepting the deliveries.

b) Orientations

The contractor is to develop a project orientation that includes but is not be limited to site-layout, safety policies, emergency protocols, and cardinal rules/golden rules.

c) Personal Protection Equipment (PPE)

All PPE is to be selected and used in accordance with legal requirements and based on appropriate job hazard analysis (JHA). PPE is to meet appropriate, recognized approval requirements (e.g., CSA, ANSI, IFC, NFPA), where applicable. The Contractor will establish minimum PPE requirements that include, at a minimum:

- i) Head protection will be climbing-style helmets, Type 2, with chin straps. The appropriate Class should be determined by a risk assessment or JHA.
- ii) Safety footwear will meet ASTM F-2412-2018 "Standard Test Methods for Foot Protection" and ASTM F-2413-2018, "Standard Specification for Performance Requirements for Protective Footwear" or provide equivalent protection and or Grade 1 CSA in Canada.
- iii) ANSI rated safety glasses with side shields and or ANSI approved visor.
- iv) Long trousers/pants.
- v) Minimum 4" sleeved shirts.
- vi) high visibility vests (Type 2) or shirts.

As determined by the JHA, hand protection and cut resistant sleeves should be worn as appropriate to address risks to hands and upper extremities.

d) Cardinal Rules/Golden Rules

The Contractor should establish rules that, if broken, may result in immediate removal from site. These rules should be communicated to all people entering the site during the site-specific safety orientation. Examples of these rules include, but are not limited to:

- i) Walking underneath a suspended load.
- ii) Entering a RED (requiring permission to enter) designated controlled access zone without permission.
- iii) Working at heights of greater than 6 feet without fall protection.
- iv) Violation of the electrical safety policy.
- v) Violation of energy isolation policy.

e) Drug and Alcohol Policy

Drug paraphernalia, controlled substances, and alcohol will be prohibited while on the work site. Workers are prohibited from reporting to the premises under the influence of drugs or alcohol.

The Contractor's site-specific safety plan is to include pre-employment drug testing and post-accident drug and alcohol testing, at a minimum.

f) Weapons

Weapons, such as guns of any kind, are not allowed on the site. They are never allowed on the person or jobsite.

g) Smoking

The Contractor, as part of the site-specific safety plan, is to establish designated smoking areas and no smoking zones on the construction site. Smoking and smokeless tobacco products are not permitted inside any STACK Infrastructure operational buildings, or near flammable liquid storage areas.

h) Stop Work Authority

The Contractor's site-specific safety plan should include a Stop Work Authority policy that enables any employee or trade partner to immediately Stop Work if a hazard is present that threatens the health or well-being of any individual. There will be no disciplinary actions allowed against workers who execute their Stop Work Authority.

i) Worker Classifications

The Contractor is to establish a worker classification system for those entering the site. Workers should be classified as permanent, phased-support, short-term, transient, or a visitor. The Contractor's system is to address how the hazards associated with the classification are addressed, including that transient and visitor personnel should always be escorted while on the construction site.

Contractors shall maintain daily logs of all people entering and exiting the project site.

j) Short-Service Employees (SSE)

The Contractor, in cooperation with trade partners, is to develop an SSE program that identifies the hazards associated with the short service scope of work and controls to mitigate significant risks.

k) Sanitation

The Contractor is to provide a designated break area as part of their site mobilization plan. The break area is to be kept clean and protected from weather exposures, including provisions for hot and cold weather seasons. Personnel should be provided with seating, tables, and an area to store lunches. The Contractor is to provide hand washing stations and trash receptacles. The Contractor is to provide adequate restroom facilities (Sites in Alberta, see Schedule 7 of Alberta Occupational Health and Safety Code), including hot and cold running water. The Contractor is to provide potable water to all personnel, including water coolers and during hot weather. Adequate shade will also be available for workers during hot weather.

l) Housekeeping

The Contractor will maintain all work areas in a tidy state, free of debris and rubbish/trash. Unless directed by the Owner, the Contractor must dispose of all debris, trash, spoil, and any hazardous waste.

In cases where an inadequate standard of housekeeping occurs and can compromise safety and cleanliness, the Owner has the right to instruct the Contractor to stop work until minimum standards have been restored. Neither additional cost charges nor extension of time to the Contract will be allowed because of such stoppage. Failure to comply may result in site cleaning by another contractor at the cost of the non-complying Contractor.

The Contractor is to carry out regular housekeeping inspections to ensure maintenance of satisfactory standards. Housekeeping crews should actively assist in creating and maintaining a safe work environment by being aware of unsafe conditions and bringing these conditions to the attention of appropriate personnel, which avoids the creation of trip hazards or potentially unsafe conditions.

m) Roads and Walkways

The Contractor is to develop and keep up to date a Traffic Management Plan (Traffic Control Plan) addressing logistical risks regarding material handling, mobile equipment, lifting operations, personnel vehicle movement, designated parking areas, delivery paths, and the allowance for emergency vehicle access. Further, the Contractor will establish personnel walkways separating workers entering and walking around the site from construction equipment and personal vehicles.

n) Protection to the Public

The Contractor, as part of their site-specific safety plan is to outline methods for protecting the public while on or in the immediate vicinity of the site. The site-specific safety plan is to address all applicable Department of Transportation (DOT) or Ministry of Transportation requirements, including federal, provincial, state, and local requirements, and the Contractor is responsible for obtaining any permits required. Additionally, the Contractor is to install, maintain, and utilize a truck wheel wash (or equivalent) for vehicles exiting the site to prevent mud and debris from damaging state or private roads and the environment.

o) Emergency Planning

Prior to commencing site activities, the Contractor will provide the Owner with both electronic and hard copies of the Contractor's Emergency Response Plan that establishes procedures for extreme weather, medical emergencies, power failures, major fires, civil unrest, workplace violence, active threats, environmental spills, or other emergency situations

The Contractor is to conduct emergency response drills including, but not limited to, fire and spill drills at least once per the duration of the project and no less than annually for fire drills. The intent is

to evaluate the effectiveness of its emergency procedures and equipment and the knowledge and proficiency of response personnel. When the site is connected to an operational Data Center the timing of such drills is to be agreed upon after consultation with the Owner and is the responsibility of the Contractor. The Contractor is to report all test/drill results to the Owner upon request.

Emergency Response Plan - In the event of, the Contractor will have a plan in place to evacuate the project site, or shelter in place if appropriate, including designated personnel to account for all workers. The Contractor is responsible for project personnel in crisis management scenarios and should coordinate with the Owner to align with the Owner's emergency response plan.

i) Alberta Specific Requirement

Before workers are sent to a work site, the contractor will ensure that arrangements are in place to transport injured or ill workers from the work site to the nearest health care facility. In addition, the following requirements apply:

- a) The contractor will ensure that an ambulance service licensed in accordance with the Ambulance Services Act is readily available to the work site when travel conditions are normal.
- b) If a worker is acutely ill or injured or needs to be accompanied during transport to a health care facility, the contractor will ensure that the worker is accompanied by at least one first aider, in addition to the operator of the transportation (this does not apply for a work site with 3 or fewer workers)

p) First Aid and Medical Care

The Contractor is responsible for ensuring that first aid, first aid supplies and medical treatment are available for workers in the event of injury. The Contractor is to have at least two qualified persons on-site who can provide first aid treatment and CPR during normal working hours. See also Section titled: Safety Implementation and Administration.

Adequate eye washing stations, plumbed or portable, will be provided and accessible.

i) Alberta Specific Requirement

- a) The number of first aiders at a work site and their qualifications and training will comply with Schedule 2, Tables 5, 6 or 7 of Alberta Occupational Health and Safety Code.
- b) At the first aid location/clinic or other designated location, the Contractor is to maintain an injury log showing all recordable, lost time, and first aid injuries. The log is to be made available to the Owner upon request. A monthly record is to be sent to the Owner as part of the reporting mechanism stated in Section 12 of this specification.
- c) First aid kit, equipment and supplies shall meet the requirements of Table 3 of Alberta Occupational Health and Safety Code.

8. **High Risk Activity (HRA) Planning**

Prior to the commencement of the work, including mobilization and site set-up activities, the Contractor will demonstrate to the satisfaction of the Owner, that the Contractor has performed a hazard identification and risk assessment for all high-risk work.

The Contractor is responsible for providing an HRA Management Plan which outlines the steps taken at the jobsite to manage HRA's. The plan applies to any task completed at the jobsite which includes or may include an HRA as defined by this document.

Ensuring effective procedures and assessment systems are in place to implement direct and alternate control of hazards and mitigate risks to as low a level as possible, and to meet all the health and safety management requirements under this specification is critical.

The following activities are considered HRAs:

- i) **Excavation and Ground Disturbance:** includes any man-made cut, cavity, trench, or depression in an earthen surface and in concrete floors and pads.
- ii) **Mobile Equipment:** Vehicles and machinery not primarily designed for transporting people or property on public roads. The equipment is designed to perform specific operations such as carry, lift, and move materials, digging, scraping, loading, or mixing.
- iii) **Lifting & Rigging:** Load handling equipment includes cranes, hoists, power operated equipment, excavators, forklifts, rough terrain equipment when used with rigging that can raise, lower, and horizontally move a suspended load. Lifting is the controlled raising or lowering of a load from one point to another. Rigging is the process of securing and preparing the load for lifting, by assessing the load's weight and dimensions, and center of gravity to determine the necessary rigging gear and attaching it to the lifting device.
- iv) **Working at Heights:** Any task where a worker could fall a 6-foot distance or any elevation where a fall hazard exists. Includes work from MEWPs.
- v) **Hot Work:** Any activity that generates or has the potential to generate heat, sparks, open flame, or explosion. Examples welding, burning, abrasive blasting, brazing, cutting, grinding, powder actuated tools, soldering.
- vi) **Control of Hazardous Energy:** Any work where there is a presence of any of the following energy sources; electrical charge or current, mechanical, hydraulic, pneumatic, chemical, thermal, gravitational/potential. This includes all electrical work.
- vii) **Confined Space:** A work area that meets three criteria; large enough for a person to enter and perform work, has limited or restricted means for entry and exit, and it is not designed for continuous human occupancy.
- viii) **Hazardous Material:** Toxic objects or substances that pose health risks. Defined by 29 CFR 1910.120, 29 1926 or 40 CFR 302 or any chemical specified in 20 CFR 1910.1200, to include gases, liquid vapors, mists, dusts, or fumes identified as causing adverse health effects in exposed employees.

a) Pre-Project Planning

HRA management begins from as early in the design stage as possible, with the Contractor participating in or coordinating the following:

- i) Safety in Design meetings to ensure feasible hazard controls are contemplated and implemented for the final design that considers hazards throughout the life of the building.
- ii) Identification of HRAs in the project master schedule.
- iii) Integration of high-level risk assessments such as Preliminary Hazard Analysis (PHA) into the master schedule.
- iv) Use of Virtual Design in Construction (VDC) for HRA management, where applicable.
- v) Management of Change (MOC) Processes. HRA management should contemplate the impact of changes to equipment, technology, procedures, or materials to ensure new hazards are not created, previously controlled hazards are not reintroduced, and existing risks are not increased when original plans and work conditions change.

b) Site Risk Management

The Contractor will ensure that all risk and hazard controls are applied in accordance with 'hierarchy of control' methodology. Elimination of hazards is the main objective. If this is not possible, prevent or minimize exposure to the risk by one or a combination of:

- i) Substitution - substituting less hazardous material, process, or equipment.
- ii) Isolation - isolating the hazard from the person or the person from the hazard.
- iii) Engineering controls- redesigning equipment or work processes.
- iv) Administrative controls.
- v) PPE- only to be used when exposure cannot be controlled by other means.

c) HRA Management Plan

The HRA Management Plan will provide for the following, at a site:

- i) coordinating the project schedule to minimize trade stacking.
- ii) developing a 3-week look ahead schedule to incorporate the definition of each HRA activity.
- iii) participating with trade partners in weekly look-ahead HRA reviews for identification of upcoming HRA activities.
- iv) receiving and reviewing all trade partners Job Hazard Analysis (JHA) three weeks prior to commencing work.

d) Task Implementation

The Contractor is to manage Pre-Task Planning (PTP). Before starting a task, routine, or repetitive work, and repeat as needed. The Contractor is to conduct a PTP meeting with each trade partner and the employees who are assigned to perform the work. Each employee is to acknowledge the hazards and control measures. If anything changes while undertaking the task, a meeting is to be held to establish how the changes will be addressed. The following areas, at a minimum, should be addressed during a PTP meeting:

- i) ensuring workers have the right tools and equipment to perform the work.
- ii) verifying the JHA for the task.
- iii) evaluating the experience level of the employee.
- iv) explaining what to do and how to do it safely.
- v) reinforce the workers' Stop Work Authority.
- vi) explaining provision and use of PPE.
- vii) evaluating and explaining the potential hazards.
- viii) inspecting the work area for potential hazards; and
- ix) inspection of tools and equipment.

9. Safe Systems of Work**a) Working at Heights**

- i) The Contractor is responsible for assessment, planning, training, implementation, and monitoring for every task that requires the use of fall protection. Fall arrest equipment, work positioning equipment, or travel restricting equipment is to be used by employees when exposed to a fall of 6 feet or more, such as at floor and roof openings, upper floor and roof edges, elevator shafts, and stairwells.
- ii) Temporary railings or appropriate fall protection systems will be put in place before accessing or working from the trailer bed.

- iii) Contractors that install fall protection systems such as guardrails or hole/opening covers are responsible for the maintenance of these systems. Before starting any work with a fall exposure, the Contractor is to develop a written fall protection plan. The fall protection plan will specify:
 - a) the fall hazards at the work site,
 - b) the fall protection system to be used at the work site,
 - c) the anchors to be used during work, including ensuring workers remain positioned to minimize swing-fall hazards and do not move beyond allowable distances from their anchor point.
 - d) that clearance distances below the work area, if applicable, have been confirmed as sufficient to prevent a worker from striking the ground or an object or level below the work area, and
 - e) the procedures used to assemble, maintain, inspect, use, and disassemble the fall protection system, where applicable.

The fall protection plan is to be written by a competent person, addressing the specific issues for the project, and including a rescue plan that includes steps to rescue the worker in addition to calling the local emergency services. If rescue agreements with outside providers are used this should be documented in writing by both parties.

The Contractor, where applicable, and its trade partners are required to provide a fall protection competent person, as defined by applicable legal standards. Prior to the use of fall protection systems, alternative methods to protect workers should be evaluated through a risk assessment. Examples of alternative methods of fall protection include guardrails and scaffolds equipped with guardrails.

b) Anchorage Systems

The Contractor is to ensure that when using Personal Fall Arrest Systems (PFAS), a competent person verifies that any anchorage being used for fall arrest will hold at least 5,000 pounds per person attached. Fall arrest and horizontal lifeline systems should be designed by a qualified person or purchased from a manufacturer qualified in the design and building of these systems. These systems should have an inspection process in place.

- i) Alberta Specific Requirement
 - a) fall arrest will hold at least 16 kilonewtons or two times the maximum arresting force in any direction in which the load may be applied or 5000lbs, whichever is greater.

c) Holes

Any opening larger than two inches in any direction and not otherwise protected by a handrail or guarding system is to be covered by a temporary hole cover. All covers used to protect the floor, roof, or wall openings can hold at least two times the maximum intended load; including any equipment in the area.

Covers should be clearly labeled "opening, do not remove" and physically secured to prevent inadvertent movement of the cover.

d) Lifts

The Contractor is to establish general guidelines for using aerial boom lifts and scissor lifts. Daily inspections should be completed prior to use and be available on the equipment during operation.

Traversing a scissor lift in the elevated position is to be avoided. If the scissor lift is being moved at height for minor repositioning, a spotter is required.

Only trained and authorized personnel are permitted to operate lifts and mobile equipment.

When using lifts, 100% tie-off (such as double lanyards) is required while personnel are climbing out of or onto the platform. 100% fall restraint is required for use in all scissor lifts, boom lifts, and aerial work platforms. Personnel are only allowed to stand on and work from the work platform. Rigging or lifting from the platform or boom is not permitted, unless attachments designed by the manufacturer are used. The manufacturer's posted capacities of the lift are not to be exceeded.

Any accessories or attachments will be manufacturer approved. No field modifications or "homemade" accessories are permitted.

Contractors must comply not only with OSHA's regulations for scissor lifts and also with the ANSI/SAIA A92.6-2006 (R2014) American National Standard for Self-Propelled Elevating Work Platforms, or the most updated version and with the manufacturer's recommendations.

e) Scaffolding

Scaffolds are only to be erected, moved, dismantled, or altered by trained personnel under the supervision of a competent person, including rolling scaffolds. Upon completing the assembly of a scaffold and before use, the competent person is to fill out a documented inspection of the system. Daily inspections should be documented on the scaffold tag(s).

f) Ladders

The Contractor is to demonstrate reasonable efforts to avoid the use of ladders and promote the "ladders last" approach by actively seeking safer alternatives for elevated work tasks. The "ladders last" approach emphasizes the use of the hierarchy of controls - Elimination>Substitution>Engineering Controls>Administrative Controls - to reduce the need to use ladders, with planning controls included in the risk assessment. If ladders are required a permit system will be in place.

Platform ladders are an accepted method when a ladder is used. The use of job-made ladders is by written exception and is prohibited within buildings level to level.

g) Frame Erection

Frame erection, concrete or steel, is primarily accomplished from aerial lifts. Climbing columns is prohibited. Erector personnel should maintain 100% tie-off in areas that have fall exposures of 6 feet or greater. Traversing horizontal beams is permitted so long as proper tie-off is maintained.

h) Electrical Safety

The Contractor and its electrical and mechanical trade partners will develop a site Electrical Safety Plan outlining project-specific safety practices to energize and de-energize new and existing electrical power systems.

The plan will establish:

- i) Compliance with the current edition of NFPA 70E.
- ii) A designated, qualified Energy Marshall to ensure the Electrical Safety Plan is implemented.
- iii) Means for locking out and tagging out electrical isolation devices for the initial startup and testing as well as for a temporary service outage of energized electrical equipment or system.
- iv) Safe electrical work conditions to prevent injury or fatality due to electrical shock or arc flash due to accidental, improper, or unintended energization.
- v) Controls to eliminate hazards associated with debris or foreign objects left behind during the installation, maintenance and testing of equipment that could result in injury or property damage during energization of equipment or systems. This program must include specific criteria for ongoing prevention strategies and a means to ensure access into equipment is

- controlled and the integrity of completed inspections is maintained.
- vi) Requirements for ground fault circuit interrupter (GFCI) protection for all cord sets, receptacles, and electrical tools and equipment connected by cord and plug that are used or available for use by employees. All GFCI receptacles should be placed at the source end of the electrical service and tested before each use.
 - vii) A process for documenting inspections of the GFCIs.
 - viii) A provision that encourages continuous improvement methods, through innovative design, tools, or installation methods.
 - ix) A process for protection of power cords and other electrical cords. This may include the use of hooks, stands or similar methods.
 - x) The Contractor is responsible for providing temporary power throughout the entire site including trailers and lighting. All electrical appliances should be inspected quarterly. Exterior temporary power equipment and cabling are to be protected by hard barricades to prevent damage from vehicle traffic. This plan will include documentation of temporary power systems and a control of change/redline system and will be presented to STACK for review.

The Contractor is to utilize the STACK Commissioning Playbook, which provides a standardized methodology for the commissioning process and is the guideline implementing the Cx process on all STACK Infrastructure projects. Cx extends through all phases of new or major renovation projects, from predesign to Owner occupancy and operation, with tasks during each phase to ensure design, construction, and operator training verification.

i) Control of Hazardous Energy – Lockout/Tagout (LOTO)

The Contractor will provide a site-specific Control of Hazardous Energy (LOTO) plan which is required for work on or near any equipment or system that produces, uses, or stores hazardous energy. The plan must cover all scenarios an organization may encounter, and include definitions, roles and responsibilities, training, and program review.

The Contractor will use Job Hazard Analysis (JHA) or an equivalent tool to address the control of hazardous energy pertaining to a specific task, piece or equipment, system or work area that has the potential for the release of hazardous energy. The JHA will address the all the following: communication plan, use of signage, controlled access zone, testing regulators with auto-stop, maximum pressure testing, off-hour work, training of workers, controls for pneumatic and hydrostatic testing, safe shut down and step-by-step re-energizations procedures, and emergency plan.

- i) Pneumatic Testing Plan
 - a) Will include signage on the ends of the line being tested to ensure all workers are aware that the line is under test.
 - b) Use engineering calculations to determine the amount of energy stored in the pipe based on its size and length to determine the work exclusion zone and when the work is to be done.
 - c) With few exceptions, pressure testing of piping systems shall be carried out using a hydrostatic method.
 - d) A permit process that requires a supervisor to sign and approve the start of and completion of the pressure test and notification of the Owner is required.
 - e) When pneumatic pressure testing is required by specification for the project and a particular system and all other means are infeasible, the General Contractor is required to have executive level review and sign off when pneumatic testing above 5 PSI must occur.
 - f) A comprehensive and detailed JHA/ Pressure Test plan must be submitted for review in all situations where pneumatic testing is to be used. Notification of the Owner is required.
- ii) Common Systems requiring Pressure Testing in Construction:

- a) Plumbing Systems; Domestic Water, sanitary sewer, industrial wastewater, and stormwater drainage.
- b) Mechanical (HVAC) Systems: Hydronic Heating/cooling systems, steam, and refrigerant piping.
- c) Fire Protection System: Sprinklers and Standpipes.
- d) Gas Systems: Natural gas and propane piping.
- e) Compressed Air Systems: Industrial/commercial air lines.
- f) Process Piping: Systems for medical gas, fuel oil, or chemicals.

j) **Mobile Equipment - Powered Industrial Equipment**

The Contractor is to establish requirements for using powered industrial equipment, including:

- i) Establishing that no worker is allowed to operate mobile equipment unless there is documented evidence that the worker has been trained and is competent.
- ii) Telehandler certification (NCCCO) is required for fixed and rotational telehandlers.
- iii) The Contractor is to ensure the manufacturer's operating instructions are kept with each piece of equipment and that mobile equipment is in safe working order.
- iv) A daily documented inspection is to be carried out by the operator and the inspection kept with the equipment and made available to the Owner upon request.

General guidelines include:

- i) Keep personnel away from moving equipment and out of pinch points.
- ii) Ensure backup alarms are operational and be audible above surrounding noise.
- iii) Riding on the machines is prohibited.
- iv) The operator's manual, applicable standards, and load charts (if applicable) are available on the machine.
- v) PPE is worn within the cab unless all doors and windows are closed, and the cab is fully enclosed within a rollover protection structure.
- vi) If a worker could be caught between a moving part of a unit of powered mobile equipment and another object, an employer will (a) restrict entry to the area by workers, or (b) require workers to maintain a clearance distance of at least 600 millimeters (2 feet) between the powered mobile equipment and the object.
- vii) All lifting equipment has the capacity clearly marked on the equipment.
 - a) Equipment attachments will be authorized by the manufacturer with supporting documentation available. Supplemental load charts and/or operating instructions are available in the machine cab.
 - b) No modifications to equipment are allowed unless authorized by the manufacturer. Proof of authorization is required to be with the equipment.
 - c) The Contractor does not allow the use of cell phones or tablets by operators or workers in proximity to mobile equipment while in use.

k) **Noise**

Establish a Noise Program for protecting personnel from hearing loss who conduct work exposed to loud occupational noise and/or extended noise exposure. The program should include risk assessments, noise monitoring, hearing protection devices, and training. Hearing protection devices shall be used in areas where noise exposure exceeds permissible exposure limits as defined by local regulations.

l) Material Handling

The Contractor, as part of their site's Traffic Management Plan, is to address logistical risks associated with material handling, such as mobile equipment movement, lifting operations, personnel and vehicle movement, designated parking areas, location of overhead power lines, delivery routes, emergency vehicle access, and emergency shelters or assembly locations.

Where manual material handling is performed, the Contractor is to establish a fifty-pound maximum manual lifting limit per worker. Workers should be encouraged to utilize two-person lifts and leverage available material handling equipment such as dollies, forklifts, pallet jacks, etc. as a first choice when doing material handling activities.

The contractor will ensure that pallets used to transport or store materials or containers are loaded, moved, stacked, arranged, and stored in a manner that does not create a danger to workers.

In an operational area or in an enclosed space, no single person is to move material (pipe, conduit, wood, etc.) that is longer than 10 feet.

m) Confined Spaces

The Contractor will evaluate spaces and designate them as confined spaces or permit required confined spaces (PRCS) in accordance with local regulations. Any space that is designated as a PRCS cannot be downgraded or declassified without the permission of the Owner.

The Contractor is to notify the Owner 21 days in advance of any planned confined space work in connection with any operational areas and follow guidelines related to operations MOP coordination. When work is completed a copy of the closed permit is to be provided to the Owner. If an external rescue provider is to be contracted, the rescue agreement is to be obtained in advance and maintained on-site by the Contractor.

n) Cranes, Lifting & Rigging

i) Crane & Lifting Plan

The Contractor is to develop a crane and lifting plan for all load handling operations and critical lifts to be conducted on-site. The plans must be submitted to the Owner in advance for review and approval.

Include in the plan and submit for review, the following:

- a) Crane Drawing; Specific equipment details, including crane serial number, capacity, configuration, obstructions inspection, ground support conditions.
- b) Load Chart; Load considerations. weight, center of gravity, radii, and configuration.
- c) Area preparation; Load handling location and path of travel, blocking/cribbing, overhead lines, and ground stability.
- d) Rigging considerations; Type, specifications, configuration, inspection, need for sling protection, diagrams.
- e) Personnel; Roles, responsibilities, qualifications and any public persons or other trade personnel affected by the lift.
 1. Crane operator qualification (per ANSI B-30.5) must be certified by the National Commission for the Certification of Crane Operators (US) or or Ministry of Advanced Education, under the Apprenticeship and Industry Training Act (Alberta), or The Ministry of Training, Colleges and Universities (Ontario).
 2. Operator documents are to include a DOT medical card or equivalent if international, employer led evaluations of the model specific operator evaluation, and mandatory testing,

3. The Contractor is to provide documentation, of operator certification prior to the crane and operator being allowed on-site.
- f) Environmental Considerations; Wind speed, storms, precipitation, power lines, in area of travel or load swing, counterweight swing area barricaded.
- g) A copy of the Annual Inspection verified as current.
- h) One year of maintenance records, for the specific crane.
- i) Copy of the Proof of Insurance, verified as current.

The plan must ensure the operator does not lift a load over personnel, operational buildings, or live electrical equipment.

ii) Critical Lift Plan

A critical lift plan is specifically required when one or more of the following applies to a planned lift:

- a) If the crane boom or load will come within 15' of overhead power lines or a lift is performed between lines.
- b) Lifts made when the load weight is more than 75% of the manufacturer's rated capacity as referenced in the load chart.
- c) Tandem lifts that are made with more than one piece of load handling equipment.
- d) Lifts over occupied structures or in tight areas.
- e) Blind lifts out of the direct view of the crane operator
- f) Lifts made over non-construction areas occupied by the public.
- g) Lifts involving multiple lift rigging assemblies or other non-routine or technically difficult rigging arrangements.
- h) Load tests
- i) When two or more cranes are set up in the same operating envelope; and
- j) Any lift the operator, lift director, supervisor or employer believes should be considered critical.

iii) Power Lines

If power lines are present determine if any part of the crane, load line, or load (including rigging and lifting accessories) could come within 20 feet of the power lines before assembling or disassembling the crane. If yes, the Contractor is required to have a power line planning meeting and develop a plan to mitigate the risks.

iv) Rigging

- a) Rigger personnel are to be identified in the lift plan and the trade partner must be able to show documentation of qualification.
- b) Rigger qualifications or certifications must not be older than 5 years since the date of training and remain in line with NCCCO certification guidelines.
- c) Free rigging, direct attachment or placement of rigging equipment onto the tines of powered industrial trucks for a below-the-tines lift, is prohibited.
- d) A Job Hazard Analysis is to be completed prior to any load handling and rigging activities. It must be reviewed by all personnel involved in the rigging / hoisting operation(s).
- e) Rigging from mobile equipment such as front-end loaders, backhoes, or similar machinery is prohibited unless the equipment is specifically designed and approved by the manufacturer for lifting operations and a manufacturer-approved lifting eye or designated lifting attachment is properly installed and used.

v) Crane Inspections

- a) On Arrival, prior to allowing crane access to field operation areas, an arrival inspection must be completed.
- b) Complete a Crane Pre-Assembly / Disassembly checklist for each crane prior to beginning any crane assembly.
- c) Ensure a Post Assembly Inspection is undertaken by a third-party inspection company upon completion of assembly to ensure that it is configured in accordance with manufacturer equipment criteria.
- d) Ensure methods to control swing radius, including establishing an exclusion zone within the swing radius, are established to protect personnel.
- e) All rigging gear is to be inspected upon arrival at the project site to ensure it meets standards.

vi) Tower Cranes

Refer to ASME B30.3

vii) Alberta Specific Requirement

These limits of approach apply to any person, equipment, machine, or part thereof that approaches a powerline or any energized electrical equipment.

Alberta – Safe limit of approach		
Phase to phase	Metres	Feet
0-750 volts Insulated or polyethylene covered conductors (1)	0.3	1
0-750 volts Bare, uninsulated	1	3
Above 750 volts Insulated conductors (1) (2)	1	3
Over 750 V to 40 kV	3	10
Over 69 kV to 72 kV	3.5	11.5
Over 138 kV to 144 kV	4	13.5
Over 230 kV to 260 kV	5	20
500 kV	7	23

(1) Conductors will be insulated or covered throughout their entire length to comply with this group.

(2) Conductors will be manufactured to rated and tested insulation levels.

o) Excavation and Trenching

The Contractor is to ensure that all excavation work is carried out under the supervision of a competent person. Excavations include any man-made cut, cavity, trench, or depression in an earthen surface and in concrete floors and pads. Barricading is to be provided around all holes or openings to prevent any person being injured because of a fall. The Contractor is to verify that all utilities are identified by public mark-out and by at least two additional utility location means prior to any intrusive work. Additional utility location means may include, but are not limited to, hydro-excavation, ground penetrating radar, drawings, and Building Information Modelling (BIM) reviews. The requirement for utility locations extends to a minimum of 50 feet from the project perimeter/footprint.

The Contractor is to report any unusual conditions found, such as underground power lines, pipelines, sewers, or inconsistent materials, immediately to the Owner and if there is a risk to personnel, to immediately stop all work until risks are addressed.

Safe access and egress to excavations and trenches is to be provided. Sides should be adequately sloped, benched, shored, or otherwise protected whenever an excavation depth exceeds 4 feet. An appropriate number of egress ladders, at least one every 25 feet, is required. Employees should be protected from both vehicles and falling loads.

All excavations must be inspected daily before work begins, after any inclement weather, and verified to be reasonably free of water. The Contractor's competent person shall declare the excavation safe for entry and document all findings in the inspection log.

p) Hazard Communications, Hazardous Materials

The Contractor will develop and implement a written hazard communication plan for the site and provide an inventory of all chemicals to Owner prior to use on-site and upon request.

- i) The plan will detail how the following will be managed:
 - a) The chemical approval process includes the designation of responsible party for review and approval of Safety Data Sheets (SDS), the criteria by which a chemical is reviewed and a written exception protocol.
 - b) Maintaining an inventory of all hazardous materials present in the workplace and posting it in a place accessible to all workers with SDS on file for each material. This includes chemicals brought onto site by their trade partners.
 - c) Training workers in handling and protection from all hazardous materials.
 - d) Labeling of chemicals with appropriate name, hazard, and warnings signs.
- ii) Flammable Liquids
 - a) All flammable liquids, including paints and oils, must be handled and stored in accordance with applicable regulations.
 - b) Flammable liquids shall be kept in a designated storage area, with 'No Smoking Within 50 Feet' signs clearly posted in and around the area.
 - c) Storage locations must be protected from vehicle and mobile-equipment traffic, and access must be limited to authorized personnel only.
 - d) When the area is not in use, it must remain secured.

q) Waste Management

The Contractor is responsible for disposing of their waste in accordance with all applicable regulatory requirements. The site-specific safety plan is to detail all environmental and waste management requirements as stipulated in this document. General requirements follow.

Storage areas for hazardous materials should be identified. Areas designated for storage of hazardous materials should have the following:

- i) two points of access and egress in the event of a release or spill.
- ii) appropriate protection from severe or inclement weather.
- iii) good ventilation and, if required based on local climatic conditions, temperature control.
- iv) appropriate separation from other project areas.
- v) the entrance to the storage area clearly indicates the storage of hazardous materials and that access is restricted.
- vi) all storage containers have appropriate labels describing the contents and nature of each container
- vii) to prevent mixing of incompatible materials.

The Contractor is to control and clean up all spills. The Contractor is to immediately notify the Owner of any chemical spills, including fuels and lubricants. Following discussions and coordination

with the Owner, the Contractor will be responsible for any regulatory agency notifications for spills exceeding threshold limits.

If hazardous waste as defined under USEPA's *Resource Conservation Recovery Act* (RCRA), Subtitle C (US) or Regulation 347 Environmental Protection Act (Ontario), Environmental Protection and Enhancement Act (Alberta) are generated as part of site operations, all elements of both federal and state RCRA regulations should be met, including waste management, storage, transport, and disposal. Upon determination that surplus hazardous materials determined to be hazardous wastes require off-site disposal, these wastes should be removed from the site for disposal by a qualified hazardous waste transporter and transported to an appropriate disposal facility for final disposition, in accordance with manufacturers' recommendations and regulatory requirements. A Contractor that generates hazardous waste is not to dispose of the waste in any Owner waste containers. The Contractor is responsible for disposing of any waste generated during project activities.

r) Traffic Management

The Contractor is to develop a site-wide Traffic Management Plan that meets the following requirements:

- i) personnel exposed to traffic wear, at a minimum, ANSI (US) or CSA (Ontario, Alberta) Class II high visibility vests or equivalent, however, there may be situations where a higher rated vest is required (e.g., night work, high traffic speed, low visibility, exposure to traffic on public roadways, etc.);
- ii) only trained and authorized flaggers control vehicular traffic.
- iii) traffic control systems maintained by the Contractor.
- iv) worker exposure activities controlled using suitable barricades.
- v) speed limit signs posted at appropriate locations at the site.
- vi) not permitting riding on equipment or in the back of pickup trucks.
- vii) use of seat belts is always required.
- viii) maintain a vehicle policy that controls unnecessary vehicle traffic inside the construction area, requires verification of driver training and that precludes the use of cellular phones while driving on the site and while operating heavy equipment.
- ix) off-road vehicles (ORVs), which includes all terrain vehicles (ATVs), utility vehicles, golf carts, and mini trucks are made readily visible and equipped with seatbelts, headlights, horn, and back up alarm; and
- x) ensure that workers using ORVs receive adequate training.
- xi) Portable steps or ladders designed for use with flatbed trailers will be used to access trailer beds during loading and unloading activities. See also Section titled Working at Heights.

The Contractor is to establish personnel walkways and provide hard barricades separating mobile equipment and vehicles from pedestrian pathways including from the construction office trailers to and around the construction site areas. The plan is to be frequently updated as conditions change. The Contractor is to provide the Owner updated versions of the plan when site conditions change.

s) Temperature extremes - Heat and Cold Stress

The Contractor is to define hazards associated with working in temperature extremes to prevent injury and temperature related illness, and is to establish protocols for treating affected employees, and to establish adequate controls.

Heat illness prevention methods include the following:

- i) Drinking water to be made available as outlined in the site sanitation plan.
- ii) All common drinking water containers to be supplied with potable water, along with individual drinking cups and a trash can. Employees should be encouraged to drink plenty of

fluids throughout the workday.

- iii) Providing adequate access to shade.
- iv) Training to be conducted as necessary to address issues associated with heat related environments including recognizing signs and symptoms of heat illness, contributing factors, prevention techniques, and emergency response.
- v) When working in hot environments, a heat stress monitoring plan to be implemented, including monitoring temperatures and implementing heat breaks and stop working when necessary.

Cold prevention methods include the following:

- i) Employees who become wet while working in cold weather should immediately change into dry clothes and seek warmer environments to prevent hypothermia.
- ii) If employees are likely to get wet while performing work in cold weather, appropriate water-resistant PPE is to be provided with blankets or an effective means to warm workers made available to prevent hypothermia.

Employees should be trained in the following:

- i) identifying the signs and symptoms of cold stress resulting in frost bite or hypothermia.
- ii) the effects of wind chill and control measures that can be taken.
- iii) clothing to be worn including hats, gloves, and masks to protect exposed skin; and
- iv) the importance of proper intake of liquids and the effects of caffeinated drinks and alcohol on the body.

t) Blasting Plan

The Contractor must provide an Explosives Site Safety Plan when explosives are stored on-site. The plan must be developed according to ANSI/ASSP A10.7 and applicable federal and state and local regulations.

The contractor must provide a comprehensive Master Blasting Plan prior to the placement of explosives on site or starting explosive-related operations.

u) Fatigue Management

The Contractor is to include means for managing fatigue, including consideration of employee fatigue during project planning, by:

arranging work schedules to provide enough rest opportunities by:

- i) considering shorter night work schedules,
- ii) limiting the number of consecutive shifts of night work, and
- iii) requiring a minimum number of hours prior to reporting for the next shift after working 12 hours or more.
- iv) implementing procedures to monitor and manage fatigue risks.
- v) training workers about the safety risks of fatigue and how to identify them.
- vi) including fatigue evaluation in root cause investigations following on-the-job incidents; and
- vii) providing a way for employees to anonymously report problematic work schedules.

10. **Standard Work Procedures**

a) Permits

The Contractor is to include procedures requiring written safety permits for the following work, at a minimum:

- i) hot work (burning and welding),
- ii) confined space entry,
- iii) excavation,
- iv) critical lifts,
- v) energized electrical work,
- vi) elevated work,
- vii) ladder use,
- viii) pressure testing,

None of the above work is to commence without a permit authorized and reviewed by the Contractor. The permits should be kept on file or available electronically in the Contractor's office and made available to the Owner upon request.

b) Silica

If workers or trade partners could be potentially exposed to respirable crystalline silica above the OSHA Action Level of $25 \mu\text{g}/\text{m}^3$, as an 8-hour Time Weighted Average (TWA) (Schedule 1 of Alberta Occupational Health and Safety Code), the Contractor should establish a written exposure control plan in accordance with all applicable regulations and those established in OSHA's revised Crystalline Silica Rule: Construction or Alberta Government's Crystalline Silica requirements.

The Contractor should have a competent person on-site that can identify existing and foreseeable respirable crystalline silica hazards in the workplace and who has authorization to take prompt corrective measures to eliminate or control them.

c) Hot Work

Any work that produces a spark or flames capable of initiating a fire or explosion is considered "Hot Work" and a permit will be obtained prior to commencing. An ABC portable fire extinguisher should be kept at each location where the work will be performed, including all portable welding machines and burning carts. The permit shall indicate the following:

- i) the nature of the hazard,
- ii) the type and frequency of atmospheric testing required, if applicable
- iii) the safe work procedures and precautionary measures necessary, and
- iv) the protective equipment required.

Prior to commencing work, a fire watch should be established to ensure that all combustibles have either been cleared from the area or are protected from any sparks. The fire watch should remain in the hot work area for at least 30 minutes after the hot work has finished to ensure that no latent fire exists. When working in an enclosed building in operational areas or in proximity to fuel storage areas, the fire watch should remain for one hour after the hot work has ceased.

d) Demolition

Before work starts on any demolition activities, the Contractor and its trade partners should develop a demolition plan and ensure that a qualified person has conducted an engineering survey for the scope of work. The engineering survey should be documented in writing and provided to the Owner for approval at least 7 days prior to starting the demolition activities. The survey should include details on methods of

removal, integrity of the structure (including surrounding structures that may be affected), energy isolation, and provisions to safeguard all personnel (to include the public, where applicable) from the hazards associated with demolition activities.

e) Power Tools

The Contractor will ensure there is a policy related to the use of hand and power tools. It should address:

- i) Limiting use to authorized and qualified personnel and ensure training specific to the tool and task.
- ii) Promoting the use of dual-triggers, anti-kickback, anti-seize devices and safety handles.
- iii) Tools must be used in accordance with the manufacturer's direction and all provided safety devices must be in place and utilized.
- iv) All power tools must include a cordless option unless the tool is unavailable or has the contractor's approval.
- v) A power tool inspection process.
- vi) Use of reciprocating saws is prohibited, unless the Contractor can demonstrate there are no alternative methods available to complete the work and an exemption is granted for the specific scope of work.
- vii) Powder Actuated Tools (PAT) are prohibited. Exemptions will be considered if adequate justification is provided to demonstrate the absence of alternative methods. The exemption process is specific to scope and activity and will include justification as to why the work or task cannot be done by other means and be accompanied by a risk assessment. Once granted a variance, the Contractor will be required to follow these guidelines:
 - a) All PATs should meet the design requirements of ANSI A10.3 (US) or CSA Z166 (Ontario), and only qualified operators who possess a qualified operator card will operate PATs.
 - b) Each PAT will have a lockable storage container.
 - c) PATs should be tested after each 1,000 fastenings as recommended by the manufacturer and inspection records kept.
 - d) PATs are not to be used in areas where flammable and/or explosive atmospheres are present.
 - e) Appropriate PPE should be worn when using PAT's.

f) Respiratory Protection

Prior to the use of respiratory protection, the Contractor should evaluate the use of engineering and administrative controls to effectively eliminate, reduce, or control the hazard to an acceptable exposure level. Where these controls are not effective, affected Contractor employees should wear appropriate respiratory protection and will be successfully fit-tested for the make, model, and size of the mask by a competent person. If respiratory protection is necessary, the Contractor or its trade partner should develop a respiratory protection plan that meets regulatory requirements.

The Contractor, and or its trade partners, should evaluate the respiratory hazards of any substance they will be using and take appropriate measures to protect their employees and other personnel in the area. The Owner may request that any operations producing a potential respiratory hazard be analyzed by a qualified industrial hygienist at the Contractor's expense.

g) Environmental Management

The Contractor should maintain equipment to prevent leaks, take appropriate measures to contain potential leaks, repair leaks promptly, and report leaks to the Owner immediately.

The Contractor is not to make any water discharges that are not permitted or otherwise approved by law. If the Contractor performs a discharge under an applicable permit or regulation, they should comply with all applicable requirements for cleanup.

The Contractor should perform work in a manner that prevents soil erosion and sedimentation, in compliance with applicable laws, regulations, permits, and any Owner requirements. If a Contractor meets the requirements for needing a Spill Prevention Control and Countermeasure Plan (SPCC), the Contractor will develop an SPCC plan and comply with all plan requirements. The plan, if needed, should be submitted to the Owner.

Spills resulting from any contractor activity should be reported to the Owner immediately. Immediate actions should be taken to safely stop the discharge, contain it, and clean it up in accordance with all applicable statutory requirements.

h) Water Mitigation

The Contractor has a Water Damage Mitigation Plan specific to the location and periodic evaluations occur with necessary adjustments to the plan. The Contractor has documented the locations of water systems, isolation valves, and connections to ensure water releases are prevented through planning and follow up. Preventative measures are identified and applied prior to execution of work. Work is communicated through a Wet Work Permit process and other means, and supervised by a competent person, including re-pressurization.

i) Fire Extinguishers

The Contractor is responsible for implementing methods for fire prevention, including the provision of fire extinguishers. Fire extinguishers should be located on the project site and inspected at least monthly by the Contractor. All extinguishers should be inspected, serviced, and maintained in accordance with applicable regulatory regulations. Monthly inspections should be conducted and documented on a tag attached to each extinguisher or by other appropriate inspection documentation systems. Fire extinguishers should be serviced annually. The number, type, and location of fire extinguishers should be determined in accordance NFPA 10 or International Fire Codes. The Contractor should ensure site personnel are trained on the use of fire extinguishers.

j) Compressed Gas:

- i) Will be properly secured and brought into buildings only as needed and removed as soon as work is completed.
- ii) Will have the safety cap in place unless equipped with a regulator when in use.
- iii) Will be secured via chain or in a welding cart unless designed to be freestanding.
- iv) Will have a three-part identification tag attached and the identification labels are not obscured.
- v) Oxygen cylinders shall be separated from fuel-gas cylinders or combustible materials by a minimum of 20 feet or separated by a 5ft fire resistant barrier with a fire-resistance rating of at least one-half hour.
- vi) Flammable gas cylinders will be stored in protected, ventilated and dry locations at least 20 feet from highly combustible materials.
- vii) When not in use with storage areas will be clearly demarcated to identify cylinder type, full or empty; and
- viii) All oxyfuel burning or welding units will be equipped with one combination check valve flashback arrester installed between the regulator and the hoses, and a second arrester between the hoses and torch.
- ix) Where required by law, workers should complete compressed gas training.

11. **Safety Implementation and Administration**

The Contractor is responsible and accountable for the safety performance of their personnel and all their trade partner. This section details the expectations for administering the Contractor's site-specific safety plan.

a) Training and Competency

Prior to the commencement of the work the Contractor should provide documentation to the Owner verifying that the Contractor's personnel are competent and have the appropriate qualifications, as required by this specification and applicable laws.

The Contractor should ensure that their trade partners are adequately trained in the type of work to be performed and have the appropriate qualifications, certifications, and are under competent supervision. Records of appropriate training and qualifications of all workers should be maintained (for example, crane operator's certificate, rigging and scaffold certificate, welding certificate, etc.) on site by the Contractor. These records should be made available to the Owner upon request.

b) Job-Wide Training Requirements

The Contractor should conduct an 'all-hands' safety meeting at least once a week. The meeting should be attended by all personnel on the job site including Project Managers, Superintendents, etc. Sign-in sheets should be completed including personnel names, company, and meeting topics.

c) Observations

The Contractor must establish an observation program to inspect and audit the construction site covering compliance issues, best practices, and positive safety behaviors. The following items should be included in the observation program, at a minimum:

- i) a weekly 'executive walk' during which the Contractor and trade partner management focus on positive employee recognition and broad safety culture principles.
- ii) recognition of outstanding performance by craft workers and trade partners through a recognition program.
- iii) establishing a compliance audit program with its trade partner ensuring all aspects of the trade partner site-specific safety plan are compliant with the Contractor providing the Owner with copies of all completed inspection and audit reports, and record results in a designated database, monthly; and
- iv) the Contractor and their trade partner establish a safety manager's site audit inspection process that is focused on the effectiveness of the site-specific safety system, which includes clear procedures for prompt corrective actions for identified hazards, tracking and verification of corrective actions until closure, including re-evaluation to ensure the fix was effective and did not create new hazards.

d) Safety Meetings

At a minimum, the Contractor should conduct a formal weekly safety meeting with all their trade partner safety managers/personnel to build safety awareness. Copies of minutes and action items arising from such meetings should be available for review by the Owner. Safety meetings should be address, at a minimum:

- i) accidents and EHS incidents.
- ii) hazardous conditions.
- iii) hazardous materials/substances.
- iv) work procedures.
- v) PPE.
- vi) housekeeping.
- vii) general safety topics.
- viii) job or work look-ahead issues; and
- ix) safety statistics.

Sign-in sheets are required for all safety meetings including pre-task planning meetings and may be periodically audited by the Owner.

e) **Safety Personnel**

The Contractor should have enough full-time and competent safety professionals on-site to ensure the work assigned by the Owner is conducted in a safe way and in accordance with their approved contractor site-specific safety plan. The Contractor should ensure that their trade partner have an appropriate safety team to manage the Contractor's site-specific safety plan requirements.

The Owner reserves the right to evaluate safety professional qualifications and verify certifications.

f) **Incident Management**

- i) The Contractor should have an accident and incident reporting system that is compatible with the Owner's incident management procedure and all applicable statutory requirements. Any incident, environmental damage, security incident, road traffic accident, or "near miss" or close call involving the Owner, the Contractor, its trade partners or any third-party personnel, property, plant or equipment, should be reported to the Owner immediately, whether injury to personnel or damage to property or equipment occurred.
- ii) A written report stating the known facts and conditions and including a preliminary assessment of the most likely consequence potential of the incident should be provided to the Owner within 12 hours. Comprehensive investigations should focus on system deficiencies, underlying root causes, with lessons learned fed back into planning and corrective action processes.
- iii) The Contractor's incident reporting system does not exempt the Contractor from providing accident reports required by statutory authorities and their own reporting mechanism. The Contractor should submit a finalized written report, including root cause(s) and corrective action(s), to the Owner within three working days of the incident.
- iv) The Contractor should notify the Owner as soon as possible but not more than eight hours of any incidents that require reporting to regulatory bodies. Serious injuries, multiple injuries, or fatalities require immediate Owner notification.
- v) Designated Health Care Providers and Injury Management - Contractors should consider the use of an on-call medic and roving medic services as integral elements of a coordinated approach with occupational health clinics, to manage the initial care of individuals who have sustained injury.
- vi) The Contractor should identify health care providers and a local occupational health clinic to be used for the treatment of work-related injuries. The contact information and a map/direction to the site should be posted and accessible to all employees and trade partners.

g) **Wellness and Ergonomics**

The Contractor should develop a wellness program, with the description and application of the program detailed in the site-specific safety plan and orientation program. The wellness program should include the following components:

- i) Daily stretch and flex program to prevent soft tissue injuries.
- ii) Ergonomic data and other programs to prevent repetitive stress injuries.
- iii) Health and wellness presentations covering various topics that are discussed periodically at the job-wide 'all hands' meeting.

h) Safety Reporting

The Contractor should complete and submit to the Owner a weekly report on agreed EHS indicators, both leading and lagging. The report should detail the following:

- i) contractors determined leading indicators which drive proactive safety efforts and identify areas for improvement before incidents occur.
- ii) work hours both for the month and the project to date.
- iii) head count for Contractor, their trade partners, and Owner vendors.
- iv) number of incidents; including near misses (highlight any potentially serious near misses), first aid cases, recordable injuries (injuries requiring care beyond first aid), environmental incidents, property damage incidents.
- v) number of fatalities.
- vi) project incident rates (incidents rates related to total hours worked, both monthly and running total).
- vii) outstanding safety performers.

12. Exemptions

Requests for exemptions or accommodation to any of the expectations presented within this document can be submitted for consideration to STACK Infrastructure's Director of Construction Safety. Requests will include an explanation of why the standard cannot be met and provide an overview of alternative controls for the hazard.

Requests will be submitted within 30 days of the project start date. If needed a review meeting will be held and final determinations will be made within 15 days of the request.

13. References

- ASME – American Society of Mechanical Engineers B30.3 Safety Standard for Tower Cranes
- ANSI - American National Standards Institute
- National Commission for the Certification of Crane Operators
- National Fire Protection Association (NFPA) Standard 70E- 2018
- 29 CFR OSHA 1926.59 - Hazard Communication
- 29 CFR 1926 Subpart K- Electrical and V - Electrical Power Transmission and Distribution
- OSHA 29 CFR 1926.1203 and .1204- Permit-Required Confined Spaces
- OSHA 1926 Subpart F- Fire Protection and Prevention
- OSHA 1926 Subpart L- Scaffolds
- OSHA 1926, Subpart P- Excavations
- Resource Conservation Recovery Act (RCRA), Subtitle C
- Ontario Occupational Health and Safety Act Reg. 632/05 (Confined Space regulation)
- Ontario Ministry of Labour Guideline No. 33 Working in Extreme Temperature Conditions.
- Alberta Occupational Health and Safety Code
- Alberta Occupational Health and Safety Act
- Alberta Occupational Health and Safety Regulations
- CAN/CSA Standard Z195 Protective Footwear
- CAN/CSA Standard Z94.1 Industrial Protective Headwear

- CAN/CSA Standard Z94.3 Eye and Face Protectors
- CAN/CSA Z462-12 Workplace Electrical Safety
- CAN/CSA Z94.2 Hearing Protection Devices
- CAN/CSA Z96.09 High Visibility Safety Apparel
- Alberta Environmental Protection and Enhancement Act
- Section 214 of Alberta Occupational Health and Safety Code - (LOTO)
- Part 19 of the Alberta Occupational Health and Safety Code (PIE)
- Part 10 of Alberta Health and Safety Code

14. **Definitions**

- Contractor: a third-party hired to provide General Contracting services at STACK Infrastructure facilities.
- OFCI: Owner Furnished Contractor Installed.
- Trade Partner: a third-party hired by the Contractor to provide supporting construction service